

LOCO-MP: Built-in Uncertainty Quantification for LOCO Feature Importance

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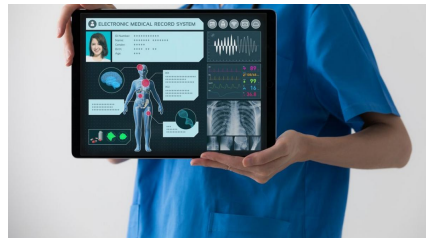
IISA

1. Motivation: Interpretable Machine Learning
2. Uncertainty Quantification for Feature Importance
 - Theoretical Guarantees
 - Simulations and Case Studies
3. Conclusion

Motivation: Interpretable Machine Learning

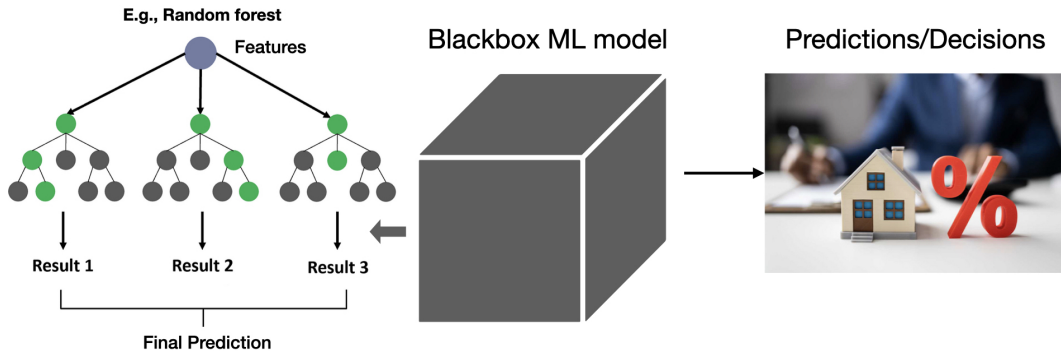
Interpreting Black-box Machine Learning Models

Machine learning is widely applied in **high-stakes applications**:



Can we trust machine learning? Make it interpretable!

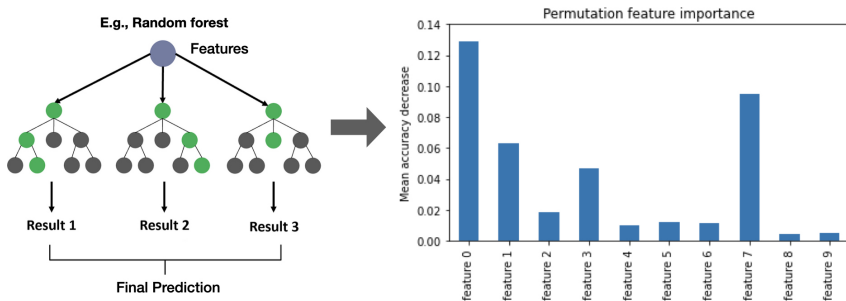
Interpreting Black-box Machine Learning Models



What is this ML system's rationale of rejecting/approving mortgage applications?

Feature Importance for Interpretable Machine Learning

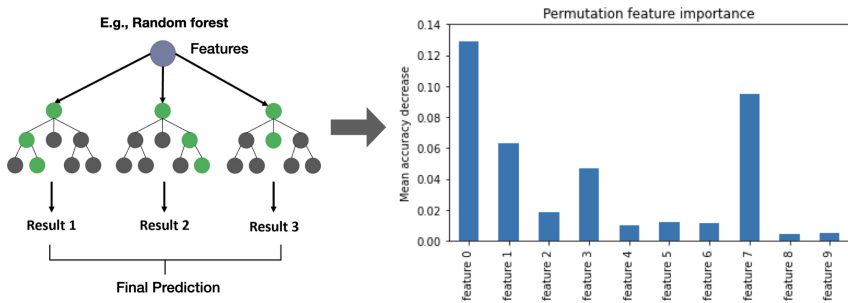
Feature importance: How does my model's prediction rely on each feature?



- Model-specific: defined for random forest, linear models, deep learning, etc.
- **Model-agnostic:** feature occlusion [Covert et al., 2021], permutation [König et al., 2021], Shapley values [Sundararajan and Najmi, 2020], etc.

Feature Importance for Interpretable Machine Learning

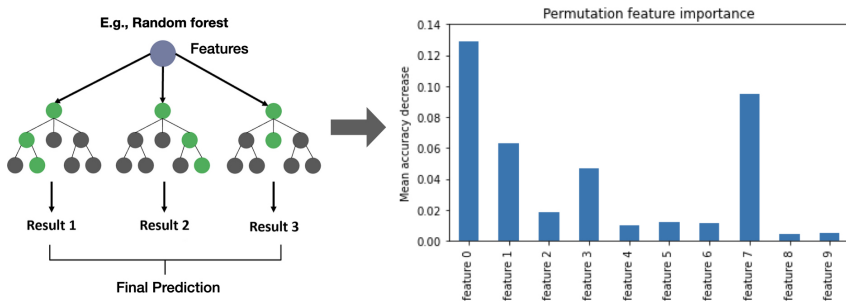
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- We focus on **global** feature importance: explanation **for the whole population** instead of one instance.
- Answers “Which features often drives the decision for most applications?”

Feature Importance for Interpretable Machine Learning

Feature importance: How does my model's prediction rely on each feature?



Many model-agnostic, global feature importance metrics exist.

Can we trust feature importance? Uncertainty quantification?

- Inference for Lasso [Lee et al., 2016, Van de Geer et al., 2014]
- Conditional independence tests for random forest [Chi et al., 2022]
- Model-agnostic methods:
Floodgate [Zhang and Janson, 2020],
GCM [Shah and Peters, 2020], VIMP
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Literature: Feature Importance Inference

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Wait a second...

- They all explain population data-generating model, not the ML model itself.
- ML models are only tools
- Impossible without strong assumptions about the data or model [Shah and Peters, 2020]!

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We call this type **population feature importance**

The feature importance we want:

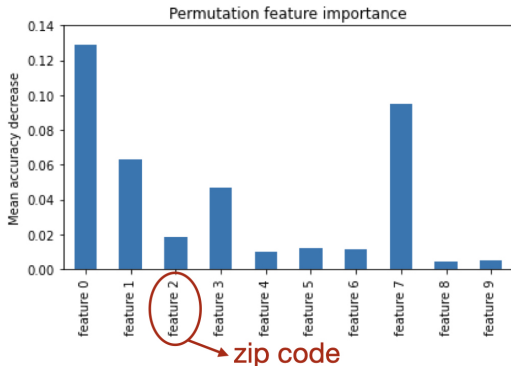
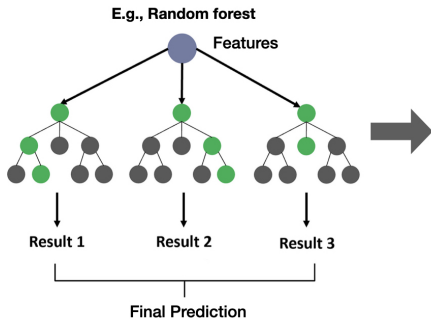
- Property of the **model**
- Which feature does my ML model rely on for decisions?
- Desired for **model diagnostics, auditing, and deployment**

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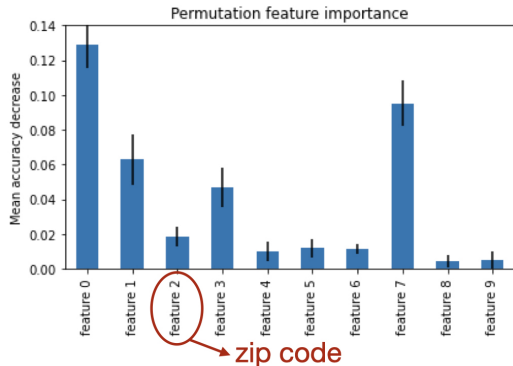
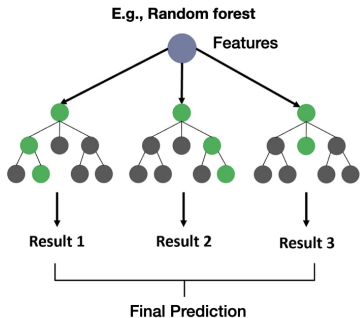
We call this type **ML feature importance**

ML Feature Importance



- How much does mortgage decision rely on sensitive features?
- E.g., zip code is a proxy of race?
- Check ML feature importance

ML Feature Importance



UQ for ML feature importance:

- has important societal consequences but is understudied!

- Only a few works [Lei et al., 2018, Rinaldo et al., 2019, Watson and Wright, 2021]
- Some require **knowing the distribution of features (model-X)** [Watson and Wright, 2021]
- The **most general, assumption-lean approach** is the **Leave-one-covariate-out (LOCO) inference** [Lei et al., 2018, Rinaldo et al., 2019].
- Practical challenges: **statistical & computational cost** for inference is substantial.

Uncertainty Quantification for Feature Importance

Leave-One-Covariate-Out (LOCO) Inference

[Lei et al., 2018, Rinaldo et al., 2019]:



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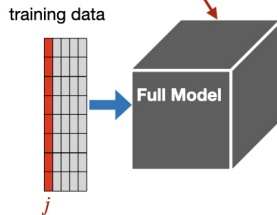


Inference target: Predictive power without feature j vs. with feature j .

Prior Work: LOCO Inference

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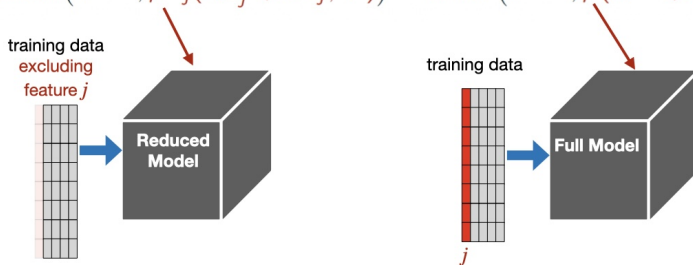
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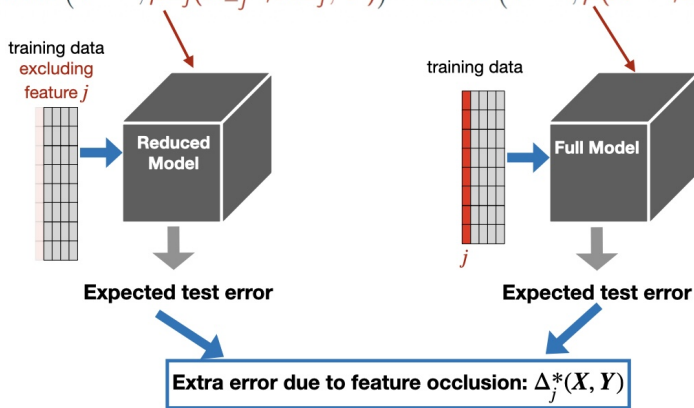
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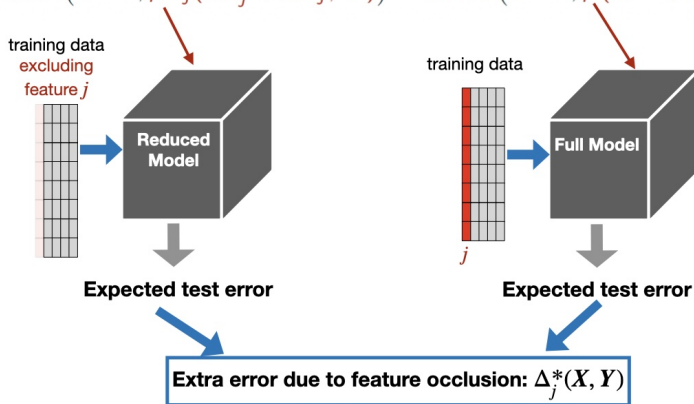
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How much does feature j **help or hurt** model μ 's predictive performance?

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- Feature occlusion scores on test data $\Rightarrow$ confidence intervals

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Challenges

- Data splitting **loses data efficiency** for both training and inference;
- Interpretation is **not for the full model** & depends on random data splitting
- Model refitting for each feature: **prohibitive computation after model training**

LOCO inference approach:

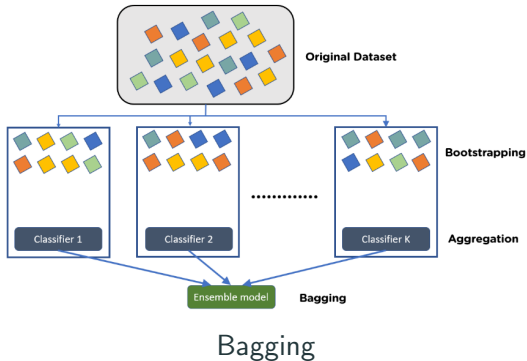
- Splits data; fits full and reduced models to training data
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Our Goal

Can we utilize the general LOCO framework to perform ML feature importance inference, while **avoiding data splitting and model refitting**?

Our Approach: LOCO Inference for an Ensemble Framework

LOCO Inference for Ensemble Learning



Picture source: <https://www.simplilearn.com/tutorials/machine-learning-tutorial/bagging-in-machine-learning>

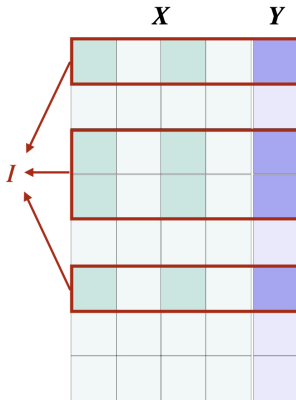
Inspiration: Jackknife+ After Bootstrap [Kim et al., 2020].

- Many ensemble methods are good predictors
- Conformal inference (Jackknife+) for bagging is **computationally free with no data-splitting!**

Idea: Minipatch Ensembles.

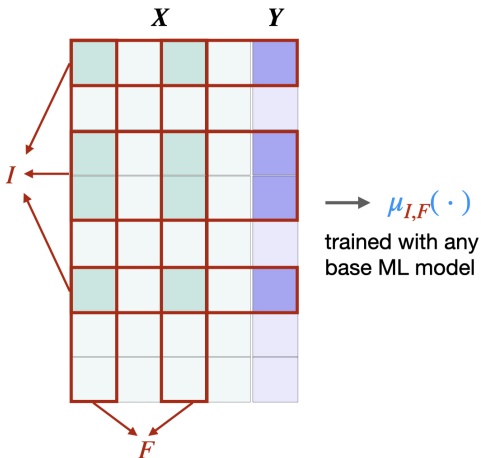
Minipatch Ensemble Learning

Minipatch ensembles: like bagging, but double-subsampling for both observations and features [Yao and Allen, 2020].



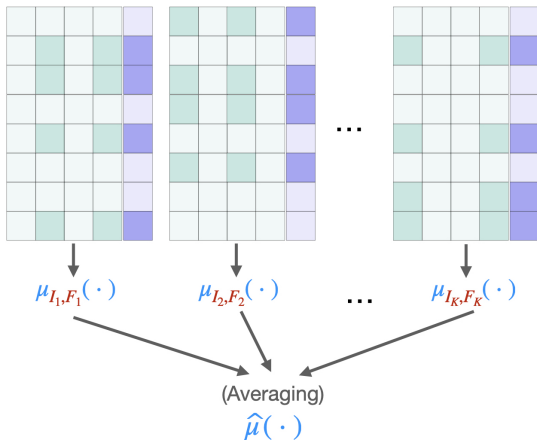
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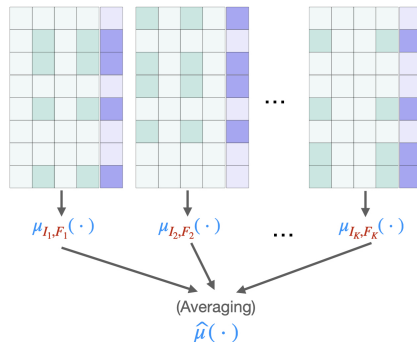


Minipatch Ensemble Learning

Inspiration: Bagging; Random Forests [Louppe and Geurts, 2012]; Stochastic Optimization & Dropout.

Advantages:

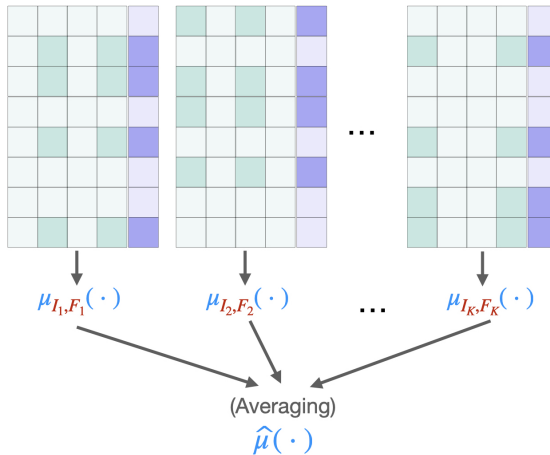
- Fast and easily parallelizable
- Ensemble diversity; **implicit regularization** [LeJeune et al., 2020, Yao et al., 2021]



LOCO Inference for Minipatch Ensembles?

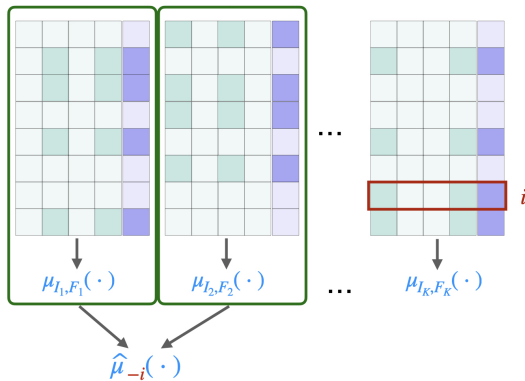
Algorithm: LOCO for Minipatch

- **Step 1.** Fit minipatch learning predictor: $\hat{\mu}$.



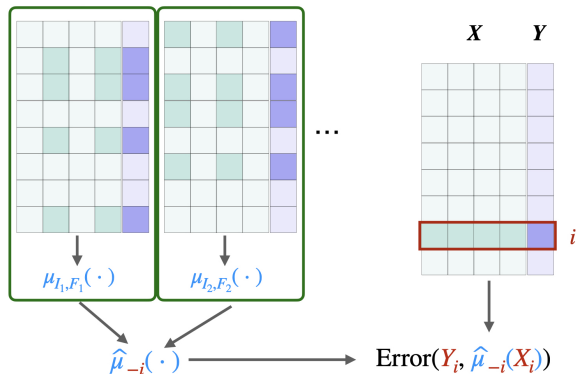
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- **Step 2. LOO** (leave-one-observation-out) predictor: $\hat{\mu}_{-i}(X_i)$.
 - Ensemble minipatches **without observation i** .
 - Compute test error on sample i .



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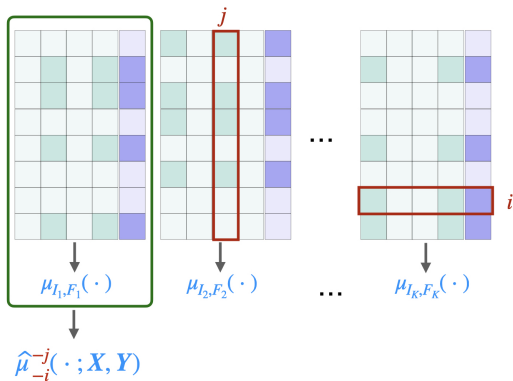
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No data-splitting!
Simple model averaging;
Free computationally!

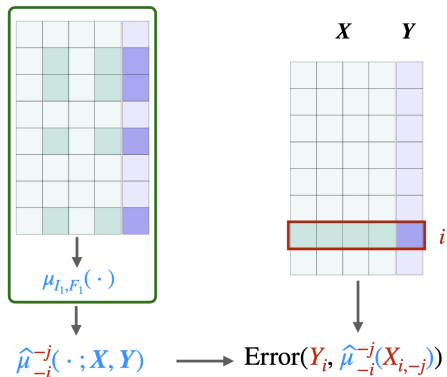
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- **Step 3. LOCO-LOO** predictor: $\hat{\mu}_{-i}^{-j}(X_i)$.
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Simple model averaging;
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Algorithm: LOCO for Minipatch

- **Step 4.** Compute feature occlusion scores for observations $1 \leq i \leq N$:

$$\hat{\Delta}_j(X_i, Y_i) = \text{Error}(Y_i, \hat{\mu}_{-j}^j(X_i)) - \text{Error}(Y_i, \hat{\mu}_{-i}(X_i)).$$

Importance of feature j for predicting sample i .

- **Step 5.** Construct asymptotically normal interval from $\{\hat{\Delta}_j(X_i, Y_i)\}_{i=1}^N$:

$$\hat{\mathbb{C}}_j = \left[\bar{\Delta}_j - \frac{z_{\alpha/2} \hat{\sigma}_j}{\sqrt{N}}, \bar{\Delta}_j + \frac{z_{\alpha/2} \hat{\sigma}_j}{\sqrt{N}} \right],$$

$\bar{\Delta}_j$: mean occlusion score, $\hat{\sigma}_j$: standard deviation.

Algorithm: LOCO for Minipatch

Full Algorithm

- **Step 1.** Fit minipatch learning predictor.
- **Step 2&3.** For each sample i , compute **LOO** and **LOCO-LOO** predictor by simple model averaging.
- **Step 4&5.** Construct the normal confidence interval.

Algorithm: LOCO for Minipatch

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Algorithmic advantages

- No **data-splitting** $\Rightarrow$ all available data is used for **training** a good predictive model and for **powerful inference**.
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-
- An ensemble framework with **built-in** LOCO inference
 - Once a minipatch ensemble is trained, **almost no extra cost (data or computation)** needed to get its own LOCO confidence intervals!
 - As a comparison: LOCO-Split require extra data set-aside for inference only, and model refitting for each feature of interest.

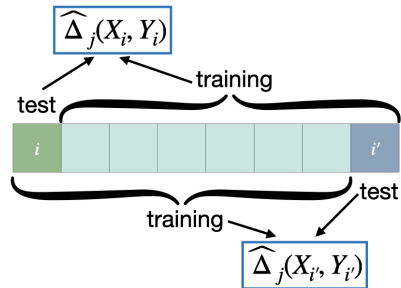
Theoretical Guarantees

Uncertainty Quantification for Feature Importance

Theoretical Guarantees

Does LOCO-MP confidence interval have valid coverage?

- **Leave-one-observation-out instead of data-splitting** $\Rightarrow$ dependency amongst $\{\hat{\Delta}_j(X_i, Y_i)\}_{i=1}^N$!
- $\hat{\Delta}_j(X_i, Y_i)$ and $\hat{\Delta}_j(X_{i'}, Y_{i'})$ switches i and i' for training and testing; **share $N - 2$ training samples**.
- Central limit theorem not applicable!



Theoretical Guarantees

- A1. Smoothness of $\text{Error}()$.
- A2. Minipatch predictors have bounded difference (automatically hold for classification).
- A3. Small MP: $n = o(\sqrt{N})$
- A4. Large number of MPs: $K \gg \frac{N}{\sigma_j^2}$

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Theorem

Suppose samples $(\mathbf{X}_i, Y_i)$ are i.i.d., and assumptions A1-A4 hold. Then

$$\lim_{N \rightarrow \infty} \mathbb{P}(\Delta_j^* \in \hat{\mathbb{C}}_j) = 1 - \alpha.$$

Valid asymptotic coverage under mild assumptions; applicable to any data distributions and base ML models.

- **Algorithmic stability**: prediction is stable against change in one training sample.
- Stability facilitates [statistical inference under dependency](#) [Bayle et al., 2020]!

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- Stability facilitates [statistical inference under dependency](#) [Bayle et al., 2020]!
- Minipatch ensembles are **stable with any base model and any data distribution!**
- Independent interest: stability also helps with conformal inference, selective inference.

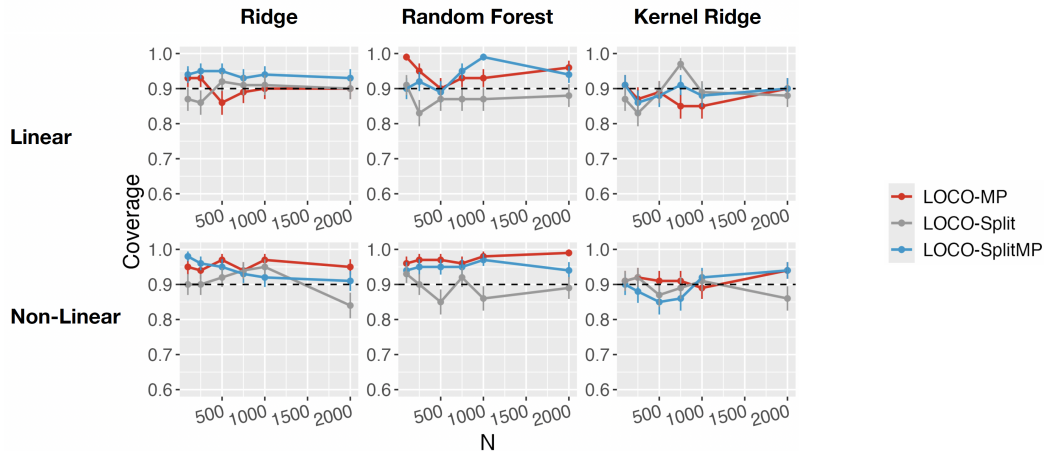
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Simulations and Case Studies

Empirical Studies

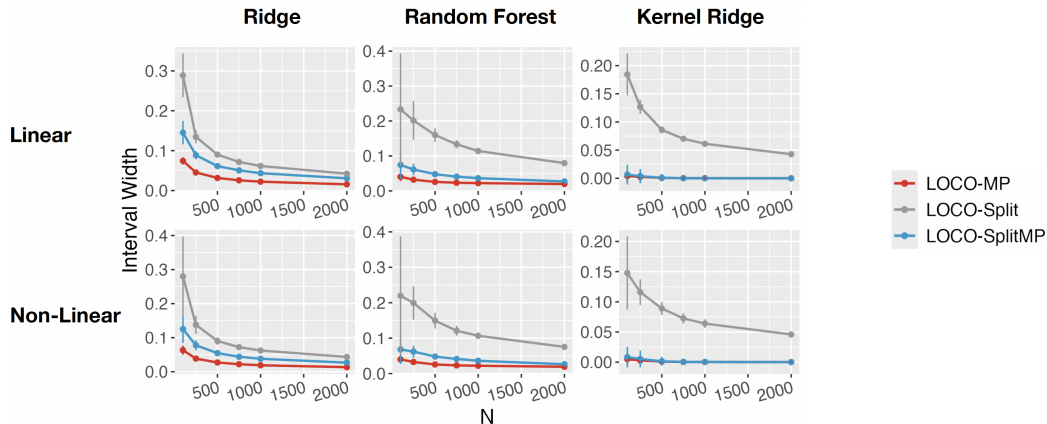
Coverage Validation

Regression, inference for a signal feature.



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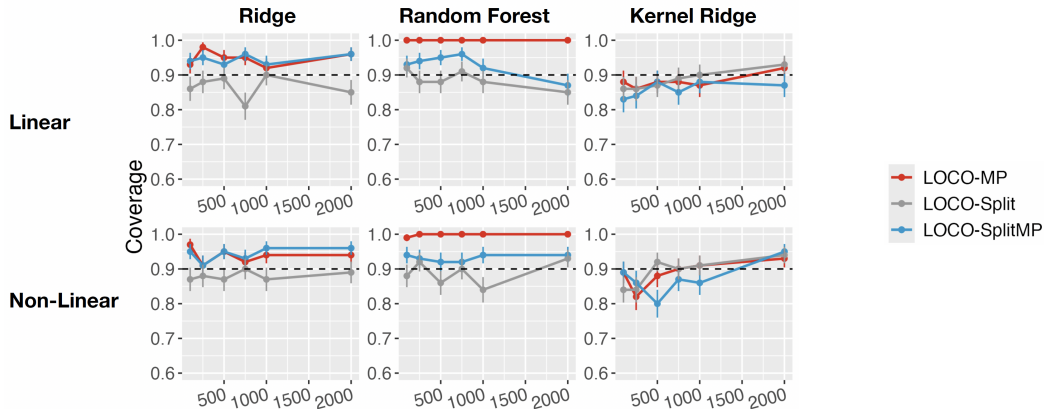
Regression, inference for a signal feature.



Width decreases as N increases; LOCO-MP has the smallest interval width.

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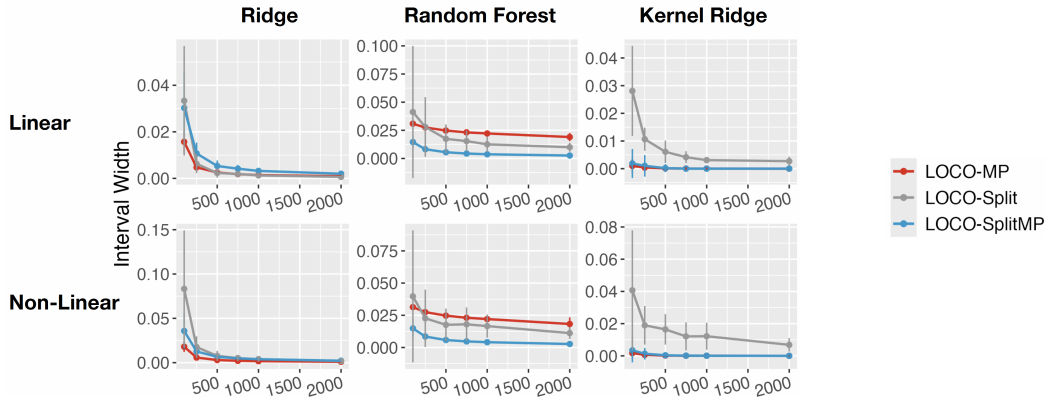
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Slight over-coverage for random forest due to variance barrier.

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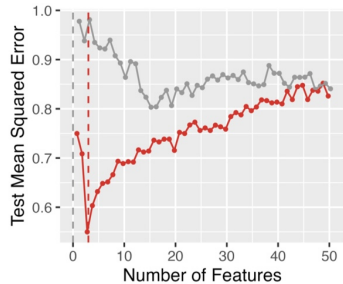
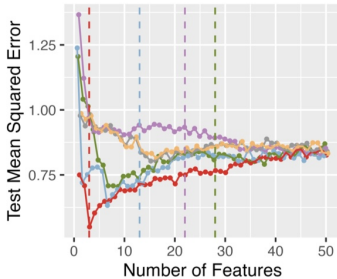
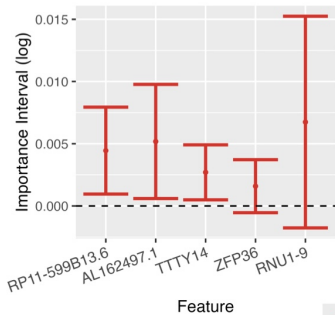


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Real Data Example

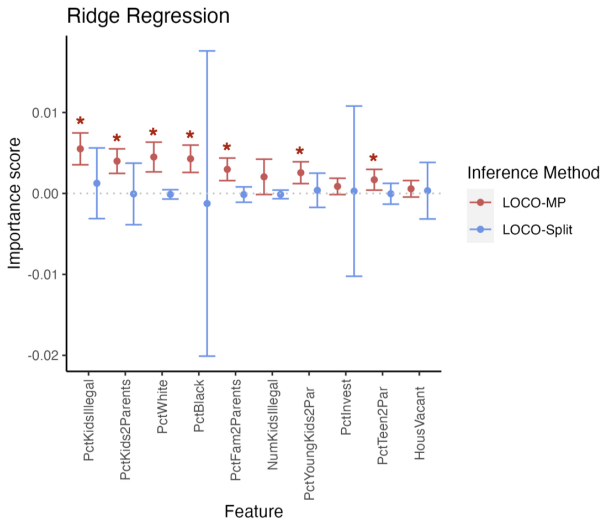
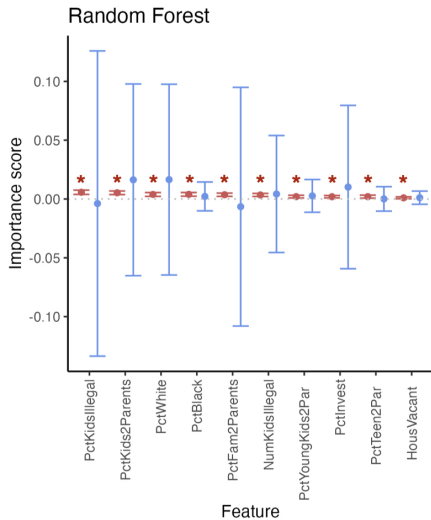
- ROSMAP data [].
- ? observations, ? features.
- Predict the Y based on X .

Real Data Example

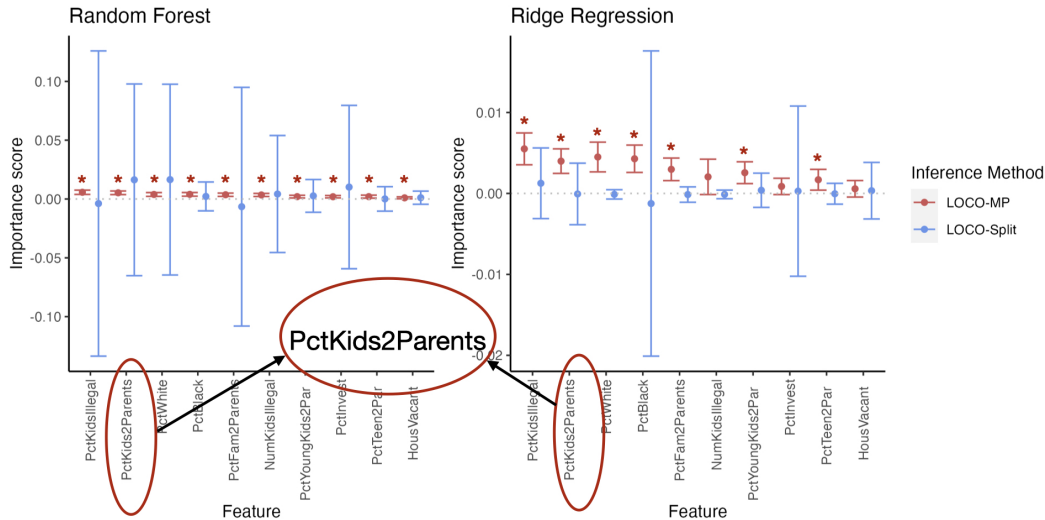


- Communities and Crimes data [Redmond, 2009].
- 1994 observations, 122 features.
- Predict the **per capita violent crime rate** based on **community features**.

Real Data Example



Real Data Example



Conclusion

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- L. Gan*, **L. Zheng***, G. I. Allen (*: equal contribution), “Model-Agnostic Confidence Intervals for Feature Importance: A Fast and Powerful Approach Using Minipatch Ensembles”, <https://arxiv.org/abs/2206.02088>.

What are minipatches?

Extension to hypothesis testing?

High dimensions: (i) multiple testing: FDR control? selective inference? (ii) are minipatch ensembles good predictors in this setting? perhaps not, with uniform sampling.

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